

NOISEMAP · DESIGN DECISION RECORD

NoiseMap TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	NSM-DDR-001	Version	0.1
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/noisemap		
Source	docs/decisions/0001-trl2-review-decisions.md @ be7ab57		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** proposed. The recommendations in items D1 to D13 are adopted for TRL 3 work pending Amish's review; items O1 to O4 remain "Proposed, awaiting Amish".

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", eight of them with a recommendation, and the design precis NSM-PRC-001 v0.2 listed its key design choices as proposed. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed the NoiseMap items one by one. Every item that carries a recommendation is therefore adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. Items without a recommendation stay open. Nothing here is recorded as decided or approved by Amish.

■ Options considered

The options for each item are those in docs/REVIEW.md (session 2026-09-25, /populate, items 1 to 9) and in NSM-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Items adopted for TRL 3 work.

#	Item	Recommendation adopted	Status
D1	Budget (review item 1)	Option (b): keep <code>budget_usd</code> at \$100 and redefine it to cover the NoiseMap parts only; the FieldNode core is costed in the FieldNode repo (\$126.00, FND-CAL-001). The full per-node cost is stated wherever deployment is discussed. R13 is restated accordingly. No new budget figure was recommended, so <code>budget_usd</code> is unchanged.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D2	Privacy claim (item 2)	State in all public copy that the guarantee rests on open head firmware with read-out protection and a published build hash; never claim privacy "by physics". NSM-CAL-001 confirms this: the cable could carry a speech codec, and the radio a few minutes of it a day.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D3	Microphone (item 3)	A small adapter board that takes either the ICS-43434 (I ² S, end of life) or the IM72D128 (PDM); the level processor accepts both interfaces.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D4	Processing location (item 4)	Levels computed in the head on a Cortex-M4F class board, so the cable carries levels only.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D5	Metrics and interval (item 5)	15-minute records with 1-minute LAeq, LAeq, LAFmax, LAFmin, L10, L90 and LCeq (22 bytes). This also settles the problem statement's questions on 1-minute detail and C weighting.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D6	Wind flagging (item 6)	Study both options at TRL 3: done in NSM-CAL-001 section G (level-based wind flag with no added cost; cup anemometer about \$25 and 0.15 kg). The choice between them is new item O3.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D7	Calibration (item 7)	A shared 94 dB, 1 kHz calibrator outside the per-node cost, and one side-by-side comparison with a class 1 meter per deployment.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D8	Public notice (item 8)	A notice on each pole saying what is measured, linking to this repository.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D9	Levels, never audio (precis choice)	No data storage and no radio in the head; firmware has no path that writes or sends samples.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D10	Standard FieldNode core (precis choice)	Standard FieldNode enclosure, 6 W panel and one cell; NoiseMap draws 12.2 mW, 12 % of FieldNode's 100 mW design allowance.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D11		Port 4.0 m above the sidewalk, 0.45 m from the pole face, facing up, on an arm with its own band clamp.	Adopted as recommended for TRL 3 under Amish's

#	Item	Recommendation adopted	Status
	Microphone position (precis choice)		2026-09-25 instruction, open for his review
D12	A and C weighting (precis choice)	LAeq and statistics, plus LCEq for amplified bass.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review
D13	Head housing (precis component 8)	3D-printed ASA tube with drip skirt and hydrophobic acoustic membrane.	Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review

Table 2. Items that remain open.

#	Item	Status
O1	First co-design partner and street (review item 9, problem statement). No recommendation was made.	Proposed, awaiting Amish
O2	R10 mass: the node is 3.32 kg against 3 kg (NSM-CAL-001 section H). Options: (a) pocket the V-blocks and saddle (about 3.10 kg, still over); (b) relax R10 to 3.5 kg; (c) both. Recommendation: (c). New at TRL 3; not applied.	Proposed, awaiting Amish
O3	R11 wind flag method: (a) level-based flag from a Z-weighted band below 40 Hz, no cost; (b) add a cup anemometer on FieldNode's second port (about \$25, 0.15 kg, which worsens O2). Rain is flagged on the server from weather data in both. Recommendation: (a), with a field comparison to set the threshold. New at TRL 3; not applied.	Proposed, awaiting Amish
O4	R12 at SF10 and slower: lengthen the interval automatically (24 min at SF10, 87 min at SF12) on The Things Network, as FieldNode proposes. New at TRL 3; follows the FieldNode decision.	Proposed, awaiting Amish

■ Consequences

- `project.yaml`: only the TRL fields change. `budget_usd` stays at \$100 (D1 redefines what it covers; no new figure was recommended). No reworded pitch or problem was recommended, so they are unchanged.
- NSM-PRB-001, NSM-PRC-001 and NSM-REQ-001 are revised to v0.3. The key design choices are no longer "proposed". R13 is redefined to cost the NoiseMap parts only, with the full node cost stated. R12's target is unchanged.
- The BOM now prices the FieldNode core at the FieldNode figure of \$126.00 (TRL 2 used \$119) and adds a street pole adapter (line 14), so the NoiseMap parts are \$62.00 and the full node \$188.00.
- NSM-CAL-001 corrects several TRL 2 figures (load 21 to 12.2 mW, autonomy 30 to 53 days, upper range 115 to 113 dBA, mass 2.3 to 3.32 kg); see its section on checks.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.